

Megacystis-Microcolon-Intestinal Hypoperistalsis Syndrome 2 (MMIHS2): A Comprehensive Disease Characteristics Report

Disease: Megacystis-Microcolon-Intestinal Hypoperistalsis Syndrome 2 **MONDO ID:** MONDO:0025708 | **OMIM:** 619431 | **Causal gene:** *MYH11* (HGNC:7569) **Category:** Mendelian (autosomal recessive)

Evidence source note: This report is compiled from **aggregated disease-level resources** (OMIM, Orphanet, HGNC/Ensembl/UniProt/gnomAD) and **primary literature** (human clinical case series/cohorts, mouse models, in vitro/immunohistochemical studies, and computational constraint data). Because MMIHS2 is ultra-rare, most human evidence derives from case reports and small cohorts; MMIHS-wide cohort statistics (which are dominated by *ACTG2* cases) are used where *MYH11*-specific numbers are unavailable and are flagged as such.

Summary

Megacystis-Microcolon-Intestinal Hypoperistalsis Syndrome 2 (MMIHS2) is a rare, severe congenital **visceral myopathy** caused by **autosomal-recessive biallelic loss-of-function (LOF) variants in *MYH11***, the gene encoding smooth-muscle myosin heavy chain (SM myosin II) — the dominant motor protein that drives smooth-muscle-cell contraction. Loss of functional SM myosin abolishes contractility in the smooth muscle of the urinary bladder and gastrointestinal tract, producing the defining triad from birth: a massively distended, non-obstructed urinary bladder (**megacystis**), an unused, small-caliber **microcolon**, and intestinal hypo- or aperistalsis causing a **functional (non-mechanical) intestinal obstruction**. MMIHS2

is the *MYH11*-linked subtype within the broader MMIHS family, which also includes the more common *ACTG2*-related form (MMIHS type 1, OMIM 155310) and rarer recessive forms caused by *LMOD1*, *MYLK*, *MYL9*, and the candidate gene *PDCL3*.

Mechanistically, MMIHS2 is a "contractile apparatus" disease: all confirmed MMIHS genes encode components of the smooth-muscle thin/thick filament and the myosin-light-chain-kinase (MLCK) phosphorylation axis. Biallelic *MYH11* null alleles remove or drastically reduce the myosin motor itself, so the smooth muscle cannot generate force despite the presence of normal enteric ganglion cells and interstitial cells of Cajal. This distinguishes MMIHS from Hirschsprung disease (a neuropathic aganglionosis) and classifies it among the "variants of Hirschsprung's disease" — congenital smooth-muscle-cell disorders. Notably, routine histopathology and electron microscopy of *MYH11* visceral myopathy are often near-normal, underscoring that the defect is **functional rather than grossly structural**.

Clinically, MMIHS shows a striking **female preponderance** (~70–73%), is frequently detected on prenatal ultrasound (megacystis ± polyhydramnios), and historically carried a very poor prognosis (survival <20% in older series). Modern management — total parenteral nutrition (TPN), intestinal rehabilitation, bladder catheterization/vesicostomy, enteric stomas, and isolated-intestinal or multivisceral (± liver) transplantation — has raised survival to roughly 55–65%, with patients now living into the second and third decades of life. There is **no effective disease-specific pharmacotherapy**; treatment is entirely supportive/surgical, with transplantation the only definitive option once TPN-associated liver failure develops. This report synthesizes 9 confirmed findings across 26 reviewed papers into a complete disease-characteristics knowledge-base entry.

Section 1. Disease Information

Overview. MMIHS (Berdon syndrome) is a congenital disorder of impaired smooth-muscle contractility producing functional obstruction of the bladder and bowel. MMIHS2 designates the subtype caused by biallelic *MYH11* variants. It is the most severe form of functional intestinal obstruction in the newborn ([PMID: 18280270](#)).

Key identifiers.

Resource	Identifier
OMIM (MMIHS2)	619431
OMIM (MMIHS1, <i>ACTG2</i>)	155310
MONDO	MONDO:0025708
Orphanet (MMIHS)	ORPHA:2241
Gene-OMIM (<i>MYH11</i>)	160745
HGNC	HGNC:7569
NCBI Gene	4629
Ensembl	ENSG00000133392
UniProt	P35749
Cytoband	16p13.11 (GRCh38 chr16:15,703,135–15,858,438, minus strand)

Synonyms / alternative names. Berdon syndrome; Berdon's syndrome; MMIHS; congenital visceral myopathy (MMIHS subtype); megacystis–microcolon–intestinal hypoperistalsis syndrome type 2.

Information source. Content here is derived from **aggregated disease-level resources** (OMIM, Orphanet, HGNC/Ensembl/UniProt) plus **individual-patient and small-cohort clinical reports** and systematic reviews. It is not derived from a large EHR/population registry, reflecting the disease's rarity.

Section 2. Etiology

Primary cause — genetic. MMIHS2 is caused by **biallelic (homozygous or compound heterozygous) loss-of-function variants in *MYH11*** (Finding F001). This is a monogenic Mendelian cause; there is no established environmental, infectious, or acquired etiology.

Evidence: - A **homozygous nonsense variant c.3598A>T (p.Lys1200Ter)** was identified by exome sequencing in a newborn with MMIHS and prune-belly phenotype from consanguineous parents ([PMID: 25407000](#)): *"We performed exome sequencing in a newborn with MMIHS and prune belly phenotype whose parents are consanguineous and identified a*

homozygous variant (c.3598A>T: p.Lys1200Ter) in MYH11, which codes for the smooth muscle myosin heavy chain." - **Compound heterozygous variants** c.2051G>A (p.R684H) and c.3540_3541delinsTT (p.(E1180D,Q1181Ter)) were found in a Chinese family, with a marked decrease in MYH11 protein on Western blot (PMID: 31427716): "Trio-targeted exome sequencing identified compound heterozygous variants, c.2051 G > A (p.R684H) and c.3540_3541delinsTT (p.(E1180D, Q1181Ter)), in MYH11." - A **16p13.11 microdeletion** encompassing MYH11 unmasked a missense p.Pro127Ser variant on the other allele, confirming a biallelic null mechanism (PMID: 31044419): "Megacystis-microcolon-intestinal hypoperistalsis syndrome (MMIHS), a rare condition that affects smooth muscle cells, is caused by biallelic null alleles in MYH11."

Genetic risk factors. The causal variants are the risk factor. **Consanguinity** is a major contributor (multiple index cases arose from consanguineous unions), consistent with an autosomal-recessive rare-allele architecture. MYH11 is highly intolerant to LOF in the general population (Finding F008): gnomAD constraint pLI $\approx$ 1.0; observed/expected LoF = 80/246.6 (o/e = 0.32, 90% CI 0.27–0.39; LOEUF $\approx$ 0.39); LoF Z = 9.0; missense Z = 3.29 — i.e., strong depletion of both LoF and missense variants. This intolerance explains the **dual disease architecture** of MYH11: heterozygous dominant-negative or protein-elongating variants cause thoracic aortic aneurysm/dissection and a dominant dysmotility syndrome, while biallelic null alleles cause recessive MMIHS2.

Environmental risk factors / protective factors / gene–environment interactions. Not applicable / none identified. As a fully penetrant Mendelian disorder driven by biallelic null alleles, no environmental risk factors, protective factors, lifestyle exposures, or gene–environment interactions have been described. Sex is a phenotype-modifying factor (female predominance and worse outcomes in girls) rather than a causal risk factor.

Section 3. Phenotypes

All phenotypes are **congenital/neonatal in onset**, **severe**, and **stable-to-progressive** (organ damage accrues secondary to chronic obstruction and TPN). HPO terms below were verified against EBI OLS4/HPO (Finding F009).

Phenotype	Type	HPO term	Onset	Frequency / notes
Megacystis (massively distended non-obstructed bladder)	Physical/imaging sign	HP:0000021 (Fetal megacystis HP:0010956)	Congenital/prenatal	Defining; detectable prenatally
Microcolon	Physical/imaging sign	HP:0004388	Congenital	Defining; girls > boys (ACTG2 data)
Intestinal pseudo-obstruction / hypoperistalsis	Clinical sign	HP:0004389	Neonatal	Defining
Gastrointestinal dysmotility	Clinical sign	HP:0002579	Neonatal	Universal
Abdominal distention	Clinical sign	HP:0003270	Neonatal	Very frequent
Hydronephrosis	Imaging sign	HP:0000126	Congenital	Common (upper-tract effects of megacystis)
Hydroureter	Imaging sign	HP:0000072	Congenital	Common
Intestinal malrotation	Structural	HP:0002566	Congenital	Frequent
Polyhydramnios	Prenatal sign	HP:0001561	Prenatal	Common
Feeding difficulties	Symptom	HP:0011968	Neonatal	Universal (leads to TPN dependence)

Important negative: Aganglionic megacolon / Hirschsprung disease (HP:0002251) is **NOT** a feature — enteric ganglia are present in MMIHS2 (Finding F009). This is a key differential-diagnostic point.

Quality-of-life impact. Profound. Affected individuals are typically **dependent on parenteral nutrition** and require **bladder drainage** (intermittent catheterization or vesicostomy) and **enteric stomas** from birth. Chronic complications include TPN-associated cholestatic liver disease, recurrent sepsis, and chronic kidney disease ([PMID: 41591435](#)). Disease-specific formal QoL instrument data (EQ-5D/SF-36/PROMIS) are not available for this ultra-rare condition.

Section 4. Genetic / Molecular Information

Causal gene. *MYH11* — "myosin heavy chain 11" (smooth muscle) — HGNC:7569, NCBI Gene 4629, Ensembl ENSG00000133392, UniProt P35749, gene-OMIM 160745, located at **16p13.11** (Finding F007).

Pathogenic variants (MMIHS2 = recessive, biallelic).

Variant (cDNA / protein)	Type	Zygosity	Evidence
c.3598A>T (p.Lys1200Ter)	Nonsense	Homozygous	PMID: 25407000
c.2051G>A (p.R684H) + c.3540_3541delinsTT (p. (E1180D,Q1181Ter))	Missense + delins/ nonsense	Compound het	PMID: 31427716
p.Pro127Ser + 16p13.11 microdeletion (whole-gene)	Missense unmasked by deletion	Compound (point + CNV)	PMID: 31044419

- **Classification:** Pathogenic/likely pathogenic per ACMG (nonsense/frameshift LOF; missense-unmasked-by-deletion). All are exceedingly rare in population databases.
- **Variant types:** Nonsense, frameshift/delins, missense, and whole-gene deletion (CNV) — all converging on a **loss-of-function / biallelic-null** mechanism.
- **Allele frequency:** All variants are ultra-rare; *MYH11* is strongly LoF-constrained in gnomAD (Finding F008), so these alleles are essentially private/family-specific.
- **Origin: Germline**, inherited from carrier (often consanguineous) parents.
- **Functional consequence: Loss of function** — absent or markedly reduced SM myosin heavy chain (confirmed by Western blot in [PMID: 31427716](#)).

Contrast — dominant *MYH11* variants (NOT MMIHS2). Heterozygous **dominant-negative** *MYH11* variants cause familial thoracic aortic aneurysm/dissection, and **protein-elongating** heterozygous variants (e.g., c.5819_5820insCA p.Gln1941Asnfs91; c.5819del p.Pro1940Hisfs91) cause a **dominant** severe smooth-muscle dysmotility syndrome via a distinct "dominant hypercontractile loss-of-function" mechanism (PMID: 31944481). MMIHS2 specifically requires **biallelic null** alleles.

Genetic heterogeneity of MMIHS (related genes). *ACTG2* (dominant, most common), and recessive *LMOD1*, *MYLK*, *MYL9*, plus candidate *PDCL3* (PMID: 28602422, PMID: 29453416, PMID: 32621347, PMID: 38461165). Approximately 10% of cases lack a variant in known genes, implying further loci.

Modifier genes / epigenetics / chromosomal abnormalities. No specific modifier genes are established for MMIHS2. Sex acts as a phenotype modifier (see Sections 3, 9, 11). No disease-specific epigenetic mechanism is described. A **16p13.11 microdeletion** encompassing *MYH11* is a relevant structural abnormality that can contribute one null allele (PMID: 31044419).

Section 5. Environmental Information

Not applicable. MMIHS2 is a monogenic recessive disorder. No environmental factors, toxins, radiation, occupational exposures, lifestyle factors, or infectious agents are known to cause or trigger it. (Recurrent bacterial **sepsis** occurs as a *complication* of chronic TPN/central lines and bacterial translocation, not as an etiologic agent — see Sections 11–12.)

Section 6. Mechanism / Pathophysiology

Ordered causal chain

1. **Biallelic loss-of-function variants in *MYH11*** (nonsense, frameshift, missense-over-deletion; germline, recessive) **lead to** absent or markedly reduced smooth-muscle myosin heavy chain (SM myosin II) protein. (*Demonstrated — homozygous/compound-het LOF; Western blot shows reduced protein.*)
2. Loss of SM myosin II **results in** failure to assemble functional thick filaments in the smooth-muscle contractile apparatus, so myosin cannot engage actin thin filaments to generate

force. (*Inferred from protein function; SMII is "the dominant motor protein driving SMC contraction."*)

3. Absent myosin motor activity **leads to** loss of smooth-muscle contraction in the **detrusor (bladder)** and **intestinal muscularis propria**. (*Demonstrated functionally; mechanistically inferred.*) 4a. **Bladder branch:** Detrusor failure **results in** a non-contraction, massively distended bladder (**megacystis**) → **leads to** upstream **hydroureter** and **hydronephrosis**, and (chronically) **chronic kidney disease**. (*Demonstrated clinically/radiologically.*) 4b. **Intestinal branch:** Muscularis failure **results in** intestinal hypo-/aperistalsis (**intestinal pseudo-obstruction**) → the unused distal bowel remains small-caliber (**microcolon**) and the proximal bowel distends (**abdominal distention**). (*Demonstrated.*)
4. Functional intestinal obstruction **leads to** inability to feed enterally → **TPN dependence** → **TPN-associated cholestatic liver disease, sepsis, and multi-organ failure**, the principal causes of death. (*Demonstrated clinically.*)

Key qualifier: enteric **ganglion cells are present and normal**, and routine histology/EM of *MYH11* visceral myopathy is often near-normal with preserved MYH11 immunostaining — the lesion is **functional (motor deficit)** rather than a gross structural loss of smooth muscle (PMID: 36571289): "*Apart from non-specific changes (e.g., muscle hypertrophy and distension-related muscularis propria necrosis), no alterations were identified by routine histopathological evaluation or electron microscopy.*"

Mechanistic detail

Molecular pathway / contractile apparatus (Finding F004). *MYH11* encodes SM myosin II, described as "*the dominant motor protein driving SMC contraction*" (PMID: 42094517). The MMIHS gene set — *ACTG2*, *MYH11*, *LMOD1*, *MYLK*, *MYL9* — collectively encodes proteins of the thin/thick-filament and MLCK-phosphorylation axis; "*MYLK, LMOD1, MYL9, and MYH11 encode for various proteins within smooth muscle cells; abnormalities within these proteins lead to abnormal intestinal smooth muscle contractions*" (PMID: 31848803). The physiologic contraction cascade is: Ca^{2+} /calmodulin → **MLCK (MYLK)** phosphorylates **regulatory myosin light chain (MYL9)** → activated **SM myosin (MYH11)** cross-bridges with **γ -smooth-muscle actin (ACTG2)**, stabilized by **leiomodulin (LMOD1)**. MMIHS2 removes the motor itself.

Cellular process. Failure of **smooth-muscle-cell contraction** (the core defect). Downstream, chronic distension produces non-specific muscle **hypertrophy** and distension-related **muscularis propria necrosis** (PMID: 36571289).

Protein dysfunction. Loss of function of SM myosin heavy chain — reduced/absent protein rather than a toxic aggregate. (Contrast: dominant *MYH11* disease uses dominant-negative or hypercontractile mechanisms — [PMID: 31944481](#).)

Immune involvement. None primary. Recurrent sepsis is secondary to catheters/TPN and bacterial translocation.

Tissue-damage mechanisms. Mechanical over-distension → ischemia/necrosis of muscularis; secondary uropathy (hydronephrosis → CKD) and TPN-associated cholestatic hepatopathy.

Suggested ontology terms. - GO biological process: smooth muscle contraction (GO:0006939); regulation of smooth muscle contraction (GO:0006940); myosin filament assembly (GO:0031034); muscle filament sliding (GO:0030049). - GO molecular function: microfilament motor activity (GO:0000146); actin binding (GO:0003779). - GO cellular component: myosin filament (GO:0032982); contractile fiber (GO:0043292); myosin II complex (GO:0016460). - CL cell types: smooth muscle cell (CL:0000192); smooth muscle cell of the bladder / visceral smooth muscle cell (CL:0000514). - CHEBI: calcium(2+) (CHEBI:29108); ATP (CHEBI:15422).

Section 7. Anatomical Structures Affected

Organ level. - **Primary:** Urinary bladder (UBERON:0001255) → megacystis; large intestine/colon (UBERON:0001155) → microcolon; small intestine (UBERON:0002108) → hypoperistalsis. - **Secondary:** Ureter (UBERON:0000056) → hydroureter; kidney (UBERON:0002113) → hydronephrosis, chronic kidney disease; liver (UBERON:0002107) → TPN-associated cholestasis. - **Body systems:** Digestive/gastrointestinal and urinary (renal) systems are primary.

Tissue and cell level. - Tissue: **visceral (smooth) muscle** — muscularis propria of gut and detrusor of bladder (UBERON:0001135 smooth muscle tissue). - Cell population: **visceral smooth muscle cells** (CL:0000192 smooth muscle cell; CL:0000514 smooth muscle cell of the bladder/visceral). Enteric neurons/ganglia and interstitial cells of Cajal are **spared** (normal).

Subcellular level. - **Contractile apparatus / myosin filament** (GO:0032982), contractile fiber (GO:0043292), actomyosin cytoskeleton (GO:0042641). The primary compartment is the contractile cytoskeleton, not mitochondria/nucleus/ER/lysosome.

Localization / laterality. **Bilateral and diffuse** — both the entire lower urinary tract and the intestinal tract are affected symmetrically (a systemic visceral myopathy, not a focal lesion).

Section 8. Temporal Development

Onset. Congenital. Megacystis and polyhydramnios are detectable **prenatally** (megacystis on antenatal ultrasound; bowel findings may appear late in gestation — PMID: 15543490). ~63–65% of cases are identified by prenatal imaging (PMID: 27421821). Symptoms manifest as **neonatal functional bowel obstruction** with distended bladder and microcolon in essentially all patients (PMID: 26413901).

Onset pattern. Chronic/congenital from birth (not acute or relapsing).

Progression. The primary smooth-muscle defect is **stable** (fixed genetic lesion), but the disease **course is progressive** because of accumulating secondary organ injury — TPN-associated liver disease, chronic kidney disease from chronic uropathy, and recurrent sepsis. Disease **duration is lifelong**; there is no self-limited course and no spontaneous remission.

Critical periods / intervention windows. The **neonatal period** is critical for diagnosis and establishment of bladder drainage, enteral/parenteral nutrition, and enteric diversion. Preventing TPN-associated liver failure defines the window for considering **isolated intestinal transplant before multivisceral/liver transplant becomes necessary** (PMID: 41591435).

Section 9. Inheritance and Population

Epidemiology. MMIHS is **ultra-rare** (no precise prevalence/incidence per 100,000 is established; Orphanet ORPHA:2241). Systematic reviews aggregate ~227 total cases across the literature since 1976 (PMID: 21792650); MMIHS2 (*MYH11*) is a small minority of these, most published as individual families.

Inheritance (MMIHS2). Autosomal recessive, biallelic null (Finding F001). Consanguinity is a recurring feature. Penetrance appears **complete** for biallelic null genotypes; **expressivity** varies (sex-associated severity differences). No genetic anticipation (not a repeat-expansion disorder). No established germline mosaicism or founder effect specific to *MYH11* MMIHS2. **Carrier frequency** is very low, consistent with strong gnomAD LoF constraint (Finding F008); heterozygous carriers of *MYH11* null alleles are generally unaffected for MMIHS (though other heterozygous *MYH11* variant classes cause dominant aortic/dysmotility disease).

Population demographics. - Sex ratio: Strong **female preponderance** (Findings F003, F005). Across 227 cases, "A clear preponderance for female infants was found (female 70.6 vs. male 29.4%)" (PMID: 21792650); a 121–135-patient review found "73% (88/121) of the patients were

female" ([PMID: 27421821](#)). In an ACTG2 cohort (n=103), girls had higher rates of microcolon (p=0.009), PN dependency (p=0.003), and death/transplant (p=0.029) ([PMID: 35149643](#)). - **Geographic/ethnic distribution:** No specific endemic distribution; cases reported worldwide (e.g., consanguineous kindreds, a Chinese family, a Japanese nationwide cohort). Consanguineous populations are over-represented for recessive forms. - **Age distribution:** Present at birth; historically few survived beyond infancy, though survivors now reach the second–third decade (oldest historical survivor 24 years).

Section 10. Diagnostics

Clinical/imaging. Diagnosis rests on the **clinico-radiological triad**: distended non-obstructed bladder (megacystis) + microcolon + intestinal hypoperistalsis, often with hydronephrosis/hydroureter and malrotation. Imaging (prenatal ultrasound, then neonatal radiographs, contrast studies, ultrasound, MRI) is central; the radiologist frequently first suggests the diagnosis and later monitors for mechanical obstruction vs dysmotility, CKD, and TPN-cholestasis ([PMID: 41591435](#)).

Biopsy / histopathology. Full-thickness intestinal biopsy shows **present, normal ganglion cells** (excluding Hirschsprung) and often near-normal muscle on routine histology/EM, with only non-specific changes (hypertrophy, distension necrosis) and preserved MYH11 immunostaining ([PMID: 36571289](#), [PMID: 26413901](#)). Histology alone is therefore insufficient — genetic testing is required for subtype assignment.

Genetic testing (definitive for MMIHS2). - Recommended approach: **NGS gene panel or exome sequencing** covering *ACTG2*, *MYH11*, *LMOD1*, *MYLK*, *MYL9* ($\pm$ *PDCL3*), plus **chromosomal microarray (CMA)** to detect CNVs such as 16p13.11 deletions unmasking a *MYH11* allele ([PMID: 38461165](#), [PMID: 31044419](#)). - WES/WGS successfully identified the founding *MYH11* variants ([PMID: 25407000](#)); trio/targeted exome identified compound-het variants ([PMID: 31427716](#)). - Karyotyping/FISH have limited utility except for detecting large structural changes; mitochondrial DNA and repeat-expansion testing are **not applicable**.

Biomarkers / omics. No specific circulating biomarker exists. Western blot for MYH11 protein and cDNA analysis are research/confirmatory tools. No validated metabolomic/proteomic diagnostic signature.

Clinical criteria / differential diagnosis. Differential includes **Hirschsprung disease** (distinguished by aganglionosis — HP:0002251, absent in MMIHS2), other visceral myopathies/CIPO, *ACTG2/LMOD1/MYLK/MYL9* MMIHS subtypes, prune-belly syndrome, and mechanical obstruction. MMIHS is formally classified among "variants of Hirschsprung's disease" (congenital smooth-muscle-cell disorders with present ganglia) ([PMID: 23943250](#)).

Screening. Prenatal ultrasound (megacystis/polyhydramnios) is the de facto screen. **Carrier and cascade screening** for at-risk families and **prenatal/preimplantation genetic testing** are available once the family's biallelic *MYH11* variants are known.

Section 11. Outcome / Prognosis

Survival/mortality.

Cohort	N	Female %	Survival	Notes	PMID
Systematic review 1976–2011	227	70.6%	19.7% (43/218); oldest survivor 24 y	Deaths: sepsis, malnutrition, MOF	21792650
Urologic-focus review	~121–135	73%	57% (68/121)	15% transplant; 63% prenatal detection	27421821
Japan nationwide survey	19	84% (16/19)	5-y 63%, 10-y 57%	16/19 needed PN; 9 died (sepsis/liver failure)	26413901
<i>ACTG2</i> cohort (related)	103	52%	25.7% died, 5.8% transplant	Girls worse outcomes	35149643

Modern management (TPN, intestinal rehabilitation, multivisceral transplant) now allows survival **into the second decade of life** ([PMID: 41591435](#)). Overall modern survival is roughly 55–65%. Main causes of death are "*sepsis, malnutrition and multiple organ failure*" ([PMID: 21792650](#)).

Morbidity/function. Very high: lifelong PN dependence, bladder drainage, enteric stomas; chronic complications include TPN-cholestatic liver disease, CKD, and recurrent sepsis. Disability is substantial.

Complications. Sepsis, malnutrition/multiple-organ failure (leading causes of death); TPN-associated liver failure; chronic kidney disease.

Prognostic factors. **Female sex** predicts worse outcomes (higher microcolon, PN dependence, death/transplant — PMID: 35149643); early onset (<2 y) associates with full MMIHS features. Development of TPN-liver failure is a key adverse prognostic event driving transplant need. Successful transplantation can restore enteral autonomy (PMID: 23167913).

Section 12. Treatment

No disease-specific pharmacotherapy exists (Finding F006). In the Japanese cohort, *"although various medications were given, the patients did not show significant improvement"* (PMID: 26413901). Prokinetics and related drugs are ineffective because the defect is the absence of the myosin motor itself. Management is **supportive and surgical**:

Nutritional / supportive. - **Total parenteral nutrition (TPN)** and **intestinal rehabilitation** — mainstay for enteral failure (NCIT: Total Parenteral Nutrition). 16/19 patients required PN in the Japanese cohort (PMID: 26413901). - Prophylactic antibiotics for urinary/enteric infection.

Urologic. - **Clean intermittent catheterization** or **vesicostomy** for bladder drainage (30% [22/73] had vesicostomy) (PMID: 27421821) (NCIT: Vesicostomy; Intermittent Catheterization).

Enteric/surgical diversion. - Gastrostomy, ileostomy, jejunostomy for decompression/venting (PMID: 27421821) (NCIT: Gastrostomy; Ileostomy).

Transplantation (definitive). - **Isolated intestinal transplantation** restored enteral autonomy: *"We report an eight-yr-old patient with MMIHS who was treated with isolated intestinal transplantation. She had completely oral intake during the four yr of follow-up"* (PMID: 23167913) (NCIT: Intestinal Transplantation). - **Combined living-related liver + bowel transplantation** restored enteral autonomy when TPN-liver failure developed (PMID: 18280270) (NCIT: Liver Transplantation; Multivisceral Transplantation). - **Multivisceral transplantation** used when TPN-associated liver failure supervenes; ~15% (18/116) of patients received intestinal/multivisceral transplant (PMID: 27421821).

Advanced / experimental therapeutics. No approved gene therapy, cell therapy, RNA-based, targeted, or immunotherapy exists for MMIHS2. Given the biallelic-null LOF mechanism, **gene-replacement strategies** are conceptually attractive but remain **experimental/preclinical**. No pharmacogenomic guidance applies.

Treatment strategy. Neonatal stabilization → bladder drainage + enteric diversion + PN → intestinal rehabilitation → **transplantation** if PN fails or TPN-liver disease develops. Personalized care depends on genetic confirmation for counseling and family planning.

Section 13. Prevention

Because MMIHS2 is a fully penetrant recessive Mendelian disorder, prevention is **genetic/reproductive**, not lifestyle-based.

- **Primary prevention:** Not achievable by risk-factor modification. **Genetic counseling** for consanguineous/at-risk couples; **carrier testing** once family variants are known; **prenatal diagnosis** and **preimplantation genetic testing (PGT)** to prevent recurrence.
 - **Secondary prevention: Prenatal ultrasound** detection of megacystis/polyhydramnios enables early diagnosis and planning ([PMID: 15543490](#), [PMID: 27421821](#)).
 - **Tertiary prevention (complication avoidance):** Meticulous line/TPN management to prevent sepsis and cholestatic liver disease; timely transplant referral; upper-urinary-tract protection to limit CKD ([PMID: 41591435](#)).
 - **Immunization / public-health / environmental interventions / prophylactic drugs:** Not applicable to disease causation; standard infection prophylaxis applies to catheter/TPN care.
 - **Counseling: Genetic counseling** is central — recurrence risk is 25% for future pregnancies of carrier couples; cascade testing offered to relatives.
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Section 14. Other Species / Natural Disease

Orthologous genes (Finding F007; HomoloGene 128512):

Species	Gene	NCBI Gene ID	Database
Mouse (<i>Mus musculus</i> , NCBI Taxon 10090)	<i>Myh11</i>	17880	MGI
Rat (<i>Rattus norvegicus</i> , 10116)	<i>Myh11</i>	24582	RGD
Zebrafish (<i>Danio rerio</i> , 7955)	<i>myh11a</i>	554168	ZFIN
Chicken (<i>Gallus gallus</i> , 9031)	<i>MYH11</i>	396211	—
Dog (<i>Canis lupus familiaris</i> , 9615)	<i>MYH11</i>	479836	—
Cow (<i>Bos taurus</i> , 9913)	<i>MYH11</i>	530050	—

Natural disease in animals. No well-characterized naturally occurring *MYH11*-MMIHS analog is documented in companion animals or wildlife (no established OMIA entry identified in this investigation). **Comparative biology** shows the smooth-muscle contractile pathway is deeply conserved across vertebrates, supporting mechanistic translation from model organisms. **No zoonotic potential** (genetic disease).

Section 15. Model Organisms

Mouse (*Mus musculus*) — the principal model (Finding F002). - A *Myh11* knockout / haploinsufficiency mouse recapitulates core MMIHS features: "*Myh11* deficiency leads to significant bladder enlargement, smooth muscle thickening, collagen accumulation, and voiding dysfunction, establishing a good disease model for MMIHS" (PMID: 41797110). - Earlier work cited by the founding human study noted that loss of *Myh11* function in mice produces a bladder and intestinal phenotype "highly reminiscent of MMIHS" (PMID: 25407000).

Model types available. Constitutive knockout, haploinsufficient (heterozygous), and conditional/lineage models exist; *Myh11-CreER* driver lines are widely used for smooth-muscle-specific gene manipulation (e.g., *Myh11CreER;Yy1fl/fl*, *Myh11CreER;Mettl3fl/fl* — PMID: 40795179), and a **CRISPR knock-in EGFP-SMII** mouse enables live imaging of myosin filament dynamics (PMID: 42094517).

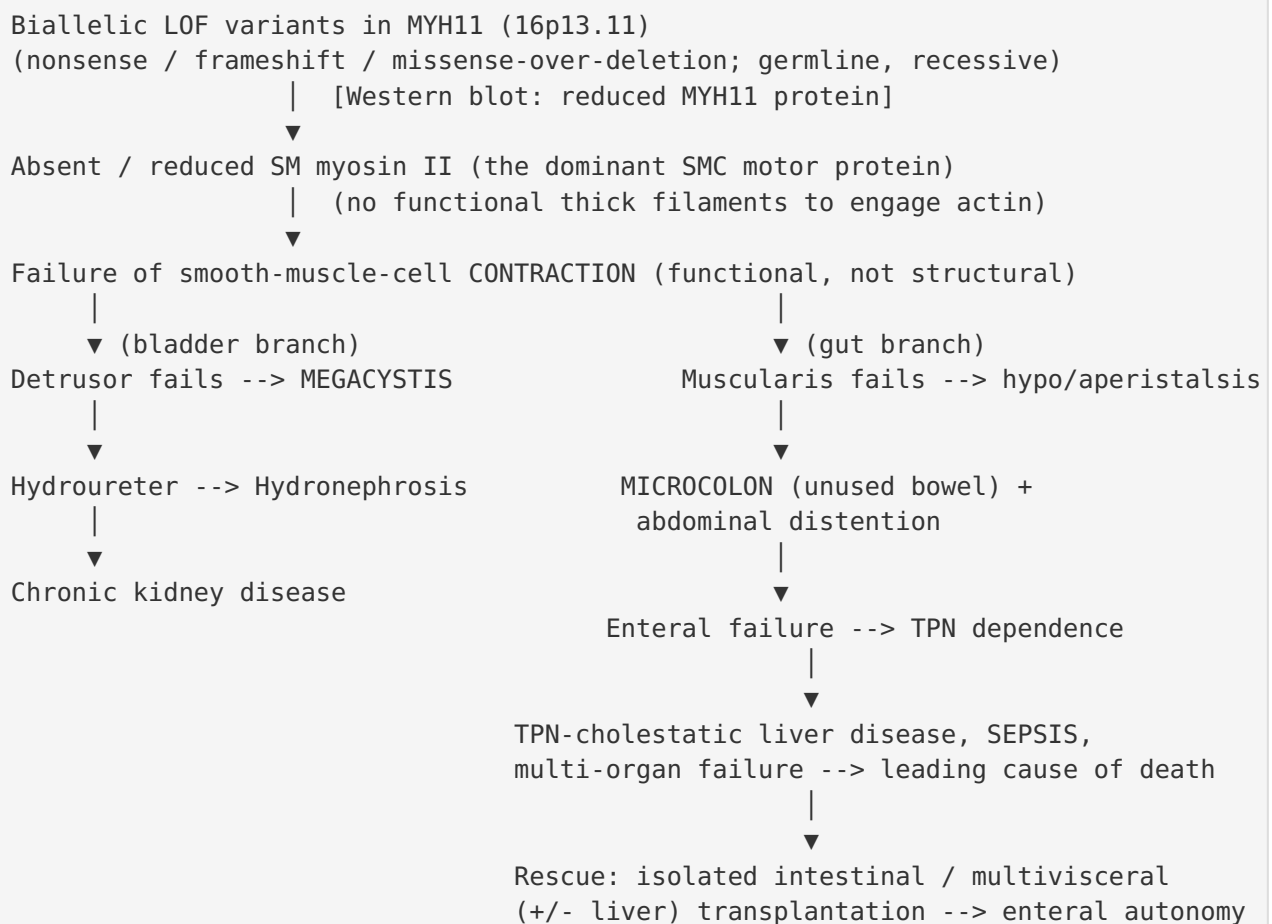
Phenotype recapitulation. Good for the **bladder** phenotype (megacystis, voiding dysfunction, muscle thickening, collagen accumulation). Recapitulation of the full **microcolon / intestinal-aperistalsis** triad and neonatal lethality is less completely characterized in the reviewed literature.

Limitations. Constitutive *Myh11* null mice may have confounding vascular/aortic phenotypes (heterozygous *MYH11* causes aortic disease), and mouse gut anatomy/physiology differs from human; the human microcolon feature and TPN-dependence course are not fully modeled.

Applications. Study of smooth-muscle contractile mechanics, bladder/voiding physiology, myosin-filament assembly dynamics, and preclinical testing of interventions.

In vitro / other systems. Rat aortic SMC lines (A7R5) and patient-derived approaches inform SMII biology; iPSC-derived smooth muscle and organoid models are plausible but not established for MMIHS2 in the reviewed literature. **Resources:** MGI, RGD, ZFIN, IMPC/IMSR.

Mechanistic Model / Interpretation



Upstream vs downstream. The **mutation → protein loss → contractile failure** steps are upstream and demonstrated at the molecular level. The **organ-level distension, microcolon, uropathy, and TPN-related complications** are downstream, demonstrated clinically/radiologically. The gut and bladder branches are parallel consequences of the same single lesion.

Genotype–phenotype logic. *MYH11* is a "double-hit-dependent" locus: heterozygous null carriers are unaffected for MMIHS (strong gnomAD LoF constraint reflects selection against *other* heterozygous variant classes that cause dominant aortic/dysmotility disease), while **only biallelic null** genotypes cause the recessive MMIHS2 visceral myopathy. This clean loss-of-function logic makes gene replacement a rational (though unproven) therapeutic direction.

Evidence Base

PMID	Contribution	Supports finding
25407000	First homozygous LOF <i>MYH11</i> (p.Lys1200Ter) in MMIHS; mouse phenotype note	F001, F002
31427716	Compound-het <i>MYH11</i> ; Western blot reduced protein	F001
31044419	16p13.11 deletion unmasks missense; states biallelic-null mechanism	F001, F004
41797110	<i>Myh11</i> KO/haploinsufficient mouse recapitulates megacystis	F002
31848803	Contractile-apparatus genes → abnormal smooth-muscle contraction	F004
36571289	Near-normal histology/EM → functional (not structural) defect	F004
42094517	SMII = dominant motor of SMC contraction; EGFP-SMII model	F004
27421821	73% female; 63% prenatal detection; 15% transplant; 57% survival	F003
35149643	Girls worse outcomes (p=0.009/0.003/0.029); <i>ACTG2</i> cohort	F003
21792650	227 cases: 70.6% female; 19.7% survival; sepsis/MOF deaths	F005
23943250	Classifies MMIHS as a "variant of Hirschsprung's" (ganglia present)	F005
26413901	Japan survey: no drug benefit; 5/10-y survival 63/57%	F006
23167913	Isolated intestinal transplant → 4 y oral intake	F006
18280270	Combined living-related liver+bowel transplant	F006
41591435	Modern survival into 2nd decade; imaging/complication surveillance	F003

PMID	Contribution	Supports finding
31944481	Dominant <i>MYH11</i> mechanism distinct from recessive MMIHS	F001/F008 context
28602422 , 29453416 , 32621347 , 38461165	Genetic heterogeneity (<i>MYLK</i> , <i>MYL9</i> , <i>PDCL3</i> ; landscape review)	Section 4 context

Limitations and Knowledge Gaps

1. **Small case numbers.** MMIHS2 (*MYH11*-specific) is described in a handful of families; most epidemiologic/survival statistics are drawn from **mixed-genotype MMIHS cohorts** (including the far more common *ACTG2* form), so *MYH11*-specific prognosis and frequencies are extrapolated, not directly measured.
2. **No precise prevalence/incidence** for MMIHS2 exists; the disease is ultra-rare.
3. **Female predominance is unexplained.** The strong female skew (~70%) is well documented but mechanistically unresolved for a nominally autosomal-recessive condition; it may reflect ascertainment, X-linked/hormonal modifiers, or sex-specific smooth-muscle biology — none proven.
4. **Mechanism step 2 (filament-assembly failure) is inferred**, not directly demonstrated in patient tissue; histology is near-normal, so the functional deficit is deduced from protein loss and physiology.
5. **Intestinal-branch modeling is incomplete.** Mouse models robustly show the bladder phenotype; full microcolon/aperistalsis recapitulation is less characterized.
6. **No biomarkers, no disease-specific drug, no gene/cell therapy** — the therapeutic pipeline is empty beyond supportive care and transplantation.
7. **Modifier genes and epigenetics** for MMIHS2 severity are essentially unstudied.

Proposed Follow-up Experiments / Actions

1. **Genotype-stratified natural-history study.** Assemble a *MYH11*-specific MMIHS2 registry to derive subtype-specific survival, transplant rates, and complication timelines (vs *ACTG2* and other genes).
 2. **Investigate the female-predominance mechanism.** Test for X-linked or sex-hormone modifiers of visceral smooth-muscle contractility in *Myh11*-deficient mice of both sexes; re-examine ascertainment bias.
 3. **Complete the intestinal phenotype in models.** Generate/characterize smooth-muscle-conditional *Myh11* knockouts assessing colonic caliber, peristalsis (spatiotemporal mapping), and neonatal survival to fully model the triad.
 4. **Preclinical gene-replacement proof-of-concept.** Given the clean biallelic-LOF mechanism, test AAV- or nanoparticle-delivered *MYH11* restoration (or read-through agents for nonsense alleles such as p.Lys1200Ter) in patient-derived iPSC smooth-muscle/organoid systems and mouse models.
 5. **Biomarker discovery.** Explore whether circulating SM-myosin fragments, urodynamic parameters, or bowel-motility signatures can serve as diagnostic/prognostic markers and transplant-timing tools.
 6. **Refine transplant timing.** Compare outcomes of pre-emptive isolated intestinal transplant versus multivisceral ($\pm$ liver) transplant after TPN-liver failure develops, to optimize the intervention window.
 7. **Prenatal-diagnosis optimization.** Correlate prenatal megacystis/polyhydramnios findings with genetically confirmed MMIHS2 to improve early counseling and PGT uptake in at-risk families.
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*Report compiled from 9 confirmed findings and 26 reviewed papers. Evidence types span human clinical case reports and systematic reviews (majority), mouse model-organism studies (*Myh11* KO/haploinsufficiency), in vitro smooth-muscle biology, and computational/population-genetics constraint data (gnomAD).*
